Monetary tightening and cumulative household purchases

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Abstract

I assess whether the first-year reduction in US consumer durable purchases following monetary tightening is offset during the next two years. I combine the original meeting-level narrative policy record with the central bank's pre-decision forecasts and direct monthly purchase volumes. The declared primary estimate is a three-year durable shortfall of 0.142 months of pre-decision purchases per 25-basis-point intended-policy innovation, with approximate 95 per cent bounds of 0.007–0.277 months. Its precision weakens under longer policy histories and leverage adjustment. A prespecified exclusion of reserves-targeting decisions increases the estimated shortfall to 0.340 months; an exploratory joint comparison gives a material sample coefficient difference whose interval excludes zero. Services also exhibit a cumulative contraction across the declared checks. These findings distinguish purchase-rate recovery from recovery of cumulative expenditure. They identify conditional responses to policy intentions under the narrative information assumptions, rather than a permanent rate-path intervention or a separately measured household timing channel.

Keywords: monetary policy; durable goods; household expenditure; cumulative responses
JEL classification: C22, E21, E52.

1 Introduction

Monetary tightening can postpone a car purchase or the replacement of household equipment. When financing conditions ease, purchases may resume, but resumption alone does not restore the expenditure lost earlier. A purchase rate that returns to its counterfactual path leaves a cumulative shortfall unless later purchases exceed that path. I ask how much of the first-year decline in US consumer durable purchases after a monetary-policy innovation is offset during the following two years. The relevant quantity is cumulative real purchases, rather than the purchase rate at a single endpoint.

The answer matters for the interpretation of monetary transmission. Durables supply services after acquisition, allowing households to postpone replacement while continuing to use their existing stock. This adjustment can concentrate an expenditure response in a relatively small part of household spending. Yet monetary policy also changes income, wealth, expected prices and debt service. A durable response is therefore not automatically an isolated intertemporal substitution effect. Comparing durables with nondurables and services helps establish the scope and timing of the response without treating those categories as an additive decomposition of real expenditure.

The declared primary estimate is a three-year durable-purchase response of -0.142 months of the pre-decision purchase flow per 25-basis-point upward innovation in intended funds-rate policy. The approximate pointwise 95 per cent interval is $[-0.277, -0.007]$. The estimate is negative in every declared comparison and every single-decision deletion, but its interval includes zero with a longer policy history or leverage-adjusted covariance. The prespecified exclusion of decision origins during the reserves-targeting episode gives -0.340 months, with bounds $[-0.548, -0.131]$. An exploratory covariance-aware follow-up gives a difference in the coefficient itself, rather than merely its significance label. Services have a smaller cumulative response that remains negative under the declared multiplicity adjustment and robustness checks.

I use the original published narrative record of Romer and Romer (2004), retaining the central bank's forecasts available before each decision. This choice gives the policy measure a documented economic meaning: an innovation in intended funds-rate policy after conditioning on the staff's inflation, growth and unemployment information. It does not equate an observed rate rise with an exogenous tightening. Monthly BEA quantities measure household purchases directly. I estimate their normalised real-flow sums, so the cumulative object does not depend on approximating volume losses by a sum of log impulse responses.

The closest economic comparison is McKay and Wieland (2021). Their model and empirical evidence already establish the importance of lumpy adjustment and subsequent durable-demand reversal. Their empirical aggregate includes residential investment, uses an extended quarterly policy series and examines much longer horizons. My contribution is a narrower monthly assessment using the original decision and forecast record, consumer-durables PCE, a fixed three-year window and jointly estimated early, later and category responses. A shortfall at month 36 cannot refute their later reversal mechanism. Neither the cumulative measure nor the partial-regression algebra used here is a theoretical novelty.

The empirical conclusions are conditional on the narrative identification assumptions and the maintained linear response model. I do not infer a household postponement duration, welfare loss or causal effect of changing the monetary operating regime from these aggregate comparisons. Section 2 explains the economic mechanism. Section 3 defines the observations and identification. Sections 4 and 5 assess the findings and robustness; Section 6 concludes. Two short appendices provide the accounting and inference details.

2 Economic mechanism and the cumulative measure

2.1 Purchase adjustment and user cost

A durable supplies services through its installed stock. For an illustrative homogeneous good, let D_t denote that stock and X_t new purchases. With depreciation $0 < \delta < 1$,

$$D_t = (1 - \delta)D_{t-1} + X_t. \tag{1}$$

The household can reduce X_t quickly while its service flow adjusts more gradually. This distinction explains why a purchase response may be large relative to the change in contemporaneous services. It does not require treating the observed BEA purchase series as a physical stock.

For a frictionless benchmark, a first-order user cost per unit of installed stock is

$$uc_t \simeq p_t^D \left(r_t + \delta - \mathbb{E}_t \pi_{t+1}^D \right), \tag{2}$$

where p_t^D is the relative durable price, r_t the real funding rate and π_{t+1}^D its subsequent relative-price growth, all measured over the same period. A higher real funding rate makes ownership more expensive. Expected resale appreciation reduces that cost. Financing constraints and fixed replacement costs make the response less smooth than this benchmark suggests. In McKay and Wieland (2021), a rate change moves households across adjustment thresholds; maintenance costs and purchases motivated by changes in household needs temper the interest sensitivity. I use this mechanism to organise the evidence, not to estimate a structural user-cost elasticity.

An intended nominal policy-rate innovation does not translate one for one into the real funding rate in equation (2). Expectations, lender margins and the subsequent policy path all matter. A contraction can also lower disposable income and asset values, reducing expenditure outside the durable category. Those general-demand effects can persist after the immediate financing response. This is why nondurables and services are informative comparisons, although they cannot separately identify income, wealth or credit channels.

2.2 What counts as recovery

Let ΔX_j be the difference between purchases following the policy innovation and purchases without it. Recovery of the purchase rate means that ΔX_j eventually returns to zero. Recovery of cumulative purchases by horizon H instead requires $\sum_{j=1}^{H} \Delta X_j = 0$. If purchases remain below the counterfactual for a year and subsequently return to it, the accumulated deficit remains. Later positive deviations are required to offset it.

Depreciation adds a further distinction. Summing equation (1) shows that cumulative purchases depend on both the endpoint stock and depreciation along the entire stock path. Even restoring the stock does not mechanically make unweighted cumulative purchases zero. Appendix A gives the short accounting argument. Consequently, my outcome measures purchase volume over a finite interval; it is not a stock, service-consumption or welfare measure.

The first twelve months and months 13–36 have unequal lengths. Their sums answer whether later spending offsets the initial reduction; their averages describe the intensity of the response. I report both interpretations explicitly. The three-year endpoint is fixed before estimation. It captures medium-term demand adjustment while leaving eventual reversal outside the claim: McKay and Wieland (2021) examine reversals emerging after roughly three to four years and extend their analysis well beyond that point.

3 Data, identification and estimation

3.1 Recorded policy intentions and pre-decision information

Romer and Romer (2004) reconstruct intended funds-rate changes from meeting records, policy documents and internal material. Intentions can include gradual implementation and some explicitly anticipated future moves. Intermeeting actions are excluded because the previous staff forecasts may not capture the information prompting them. The source then regresses the intended change on the existing intended rate, four horizons of inflation and real-growth forecasts, their corresponding revisions and an unemployment nowcast. Revisions compare forecasts for the same calendar quarter. These are genuinely pre-meeting inputs, including real-time assessments of the preceding quarter.

The original workbook contains 272 decision rows and 263 complete forecast-purge observations, March 1969–December 1996. Nine rows lack a defined purge and are not converted into zero innovations. I reproduce the published 19-column source regression from the unchanged author workbook. I retain the original series rather than splice a later extension into it. Earlier monthly zeros outside the narrative coverage are placeholders, not measured policy inaction.

My common estimation sample begins in January 1971 and ends in December 1996, with 238 eligible decisions in 237 months over a 312-month calendar. This starting point allows both the twelve-month baseline policy history and the twenty-four-month comparison to use observed intentions from January 1969 onwards. Each decision retains its own pre-meeting forecast vector. I do not aggregate a later meeting's forecasts into controls for an earlier decision in the same month. The observed source record contains two included decisions in June 1971; their common monthly outcomes are treated as dependent.

A causal interpretation requires the forecast and state controls to capture information that jointly affects policy choices and future purchases. Useful policy information omitted from the staff forecasts can violate this requirement, as Romer and Romer (2004) explicitly recognise. The dynamic response also requires that the residual innovation is not a proxy for other shocks over the projection horizon; contemporaneous orthogonality alone is insufficient (Stock and Watson, 2018). The linear specification imposes symmetric marginal responses to easing and tightening. These conditions identify a response to the policy-intentions innovation and its ensuing policy path, not a permanent externally held rate increase.

3.2 Household quantity measures and timing

I use monthly, seasonally adjusted BEA chain quantity indexes for consumer durables, nondurables and services, together with real disposable income and the PCE price index. The retained data vintage is 11

October 2026 (BEA and Federal Reserve Board, 2026). It is distinct from the real-time staff forecasts used for identification. PCE covers purchases by and on behalf of resident persons, including nonprofit institutions and imputations (BEA, 2024). Services therefore include more than cash outlays. Consumer durables include motor vehicles and household equipment; residential investment is excluded.

For category i, decision month t and horizon H, I construct

$$Y^i_{t,H} = \sum_{j=1}^{H} \frac{Q^i_{t+j}}{Q^i_{t-1}} - H, \qquad H \in \{12, 36\}, \tag{3}$$

where Q^i is the real quantity index. The next full calendar month is the first included purchase month. The denominator is the last full pre-decision month, so it cannot contain purchases made after that decision. The reference-year nominal anchor and annual-rate conversion cancel within this ratio. Subtracting H only centres the units; an intercept absorbs it, and the no-shock path can grow.

The coefficient on equation (3) is in months of the pre-decision purchase flow. A value of -0.12 at $H = 12$, for example, corresponds to an average reduction of one per cent of that baseline flow over the year. It is not an endpoint percentage change. Chained category quantities are not generally additive across categories: I compare their proportional responses without constructing an artificial total-volume aggregate.

I also estimate monthly purchase-level responses $100(Q^i_{t+h}/Q^i_{t-1} - 1)$ for $h = 1, \ldots, 36$. Their sum divided by 100 equals the cumulative outcome exactly. These flow responses are percentages of the pre-decision flow, rather than of the growing future counterfactual. Months 13–36 equal the full thirty-six-month outcome less the first twelve months on the same rows.

3.3 Matched projections and uncertainty

Let z_e be the source innovation at decision e, divided by 0.25 percentage points. For each outcome I estimate the local projection

$$Y^i_{t(e),H} = \beta^i_H z_e + W'_e \gamma^i_H + \varepsilon^i_{e,H}. \tag{4}$$

The coefficient is the response per 25 basis points of intended-policy innovation. Local projections provide direct horizon-specific response estimates (Jordà, 2005). The common conditioning vector contains all nineteen source-purge columns, preceding-month monthly and annual growth of the three purchase categories and real disposable income, annual PCE inflation, the effective funds-rate level and twelve lags of monthly raw policy intentions. The source forecasts provide the policy-information controls; the additional histories describe expenditure dynamics and past policy.

Including every current source-purge column has a useful consequence. Residualising the generated innovation on W_e gives exactly the same regressor as residualising the raw intended change. The estimated current purge coefficients therefore do not affect the matched projection coefficient or its joint score covariance. Appendix B proves this application of established partial-regression algebra (Ding, 2021). It removes a particular generated-coefficient nuisance without adding identification or correcting forecast and narrative measurement errors.

I estimate early, later and total responses jointly through their common regression scores. Same-month decision scores are summed before applying a Bartlett calendar covariance estimator (Newey and West, 1987); actual month distances are retained across gaps in the decision calendar. The declared bandwidth is 41 months: thirty-five months of overlapping purchase flows plus a six-month dependence buffer. The finite correction is $n/(n-k)$. This gives approximate normal inference, not exact finite-sample coverage. Longer histories, alternative bandwidths and leverage adjustment assess its practical sensitivity.

Smooth local projections link horizons through penalised spline coefficients, trading variance against bias and complicating confidence-interval construction (Barnichon and Brownlees, 2019). I retain the declared separate horizon regressions and their shared covariance for the cumulative purchase measure.

The single primary endpoint, durable $H = 36$, receives a two-sided 95 per cent interval. A separate nine-outcome Bonferroni family covers the three windows in each of the three categories. Each category's full profile has its own thirty-six-endpoint family; no common 108-endpoint coverage claim is made. The

primary specification, including these rules and the sample, was committed before any US spending fit. The later sample-contrast diagnostics were recorded separately after observing the initial results.

4 Cumulative household purchase responses

4.1 An initial reduction followed by further shortfall

The primary durable estimate is −0.142 months of baseline purchases over three years. Dividing by 36 implies an average purchase reduction of 0.395 per cent of the pre-decision flow for a 25-basis-point upward policy-intentions innovation. The first-year estimate is −0.036 months, or an average reduction of 0.299 per cent. The following two years contribute a further −0.106 months, an average reduction of 0.443 per cent, rather than a positive offset. The point estimates therefore show an accumulated deficit, not a purchase rebound that repays the first-year reduction.

Figure 1 makes the distinction visible. The full-sample purchase-flow point response is near zero at month 36, while its cumulative point response remains negative. The intervals do not establish exact recovery of the flow. More importantly, even a precisely zero endpoint flow would not establish recovery of cumulative purchases. The same distinction applies to the decisions outside the reserves-targeting episode, whose estimated purchase reduction remains more persistent.

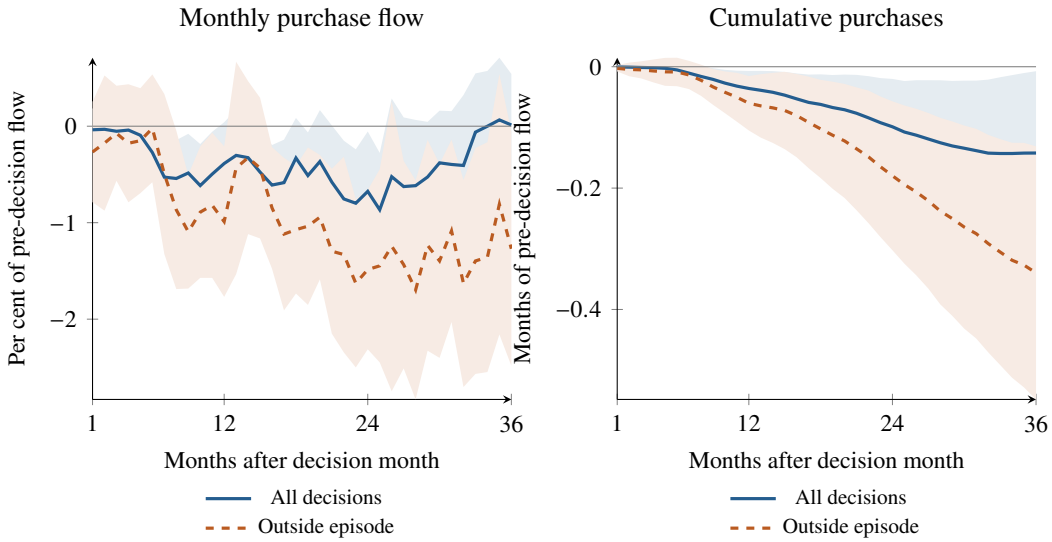


Figure 1: Durable purchase flows and their cumulative response. Both panels use a 25-basis-point intended-policy innovation. Shading gives approximate pointwise 95 per cent intervals. The right panel integrates real flows, not log changes; returning to zero in the left panel does not eliminate a cumulative shortfall. Profile-wide bounds are retained separately.

The pointwise primary interval excludes zero, but the nine-window family interval for durable $H = 36$ includes it. I therefore separate the declared primary inference from broader comparisons in Table 1. The later-period full-sample durable response also has a pointwise interval including zero. The data favour a cumulative shortfall in this specification, but do not establish its size with equal precision under every reasonable dynamic and covariance choice. Section 5 reports those choices directly.

4.2 What the other categories show

Nondurables and services have small first-year point responses and larger accumulated reductions in months 13–36. Their full-sample three-year point estimates are −0.036 and −0.038 months, respectively. Services retain a negative total and later-period response under the nine-window correction. The services total also remains negative at that correction in every declared sensitivity, including the longer

Table 1: Cumulative purchase responses in jointly estimated windows

	Months 1–12	Months 13–36	Months 1–36
All eligible decisions			
Durables	−0.036	−0.106	−0.142
95% family bounds	[−0.077, 0.005]	[−0.266, 0.053]	[−0.333, 0.049]
Nondurables	−0.005	−0.031	−0.036
95% family bounds	[−0.016, 0.006]	[−0.062, 0.001]	[−0.075, 0.004]
Services	−0.006	−0.032	−0.038
95% family bounds	[−0.015, 0.003]	[−0.053, −0.010]	[−0.064, −0.011]
Decisions outside the reserves-targeting episode			
Durables	−0.060	−0.280	−0.340
95% family bounds	[−0.124, 0.003]	[−0.526, −0.033]	[−0.635, −0.045]
Nondurables	−0.002	−0.054	−0.056
95% family bounds	[−0.031, 0.027]	[−0.133, 0.024]	[−0.153, 0.040]
Services	−0.005	−0.051	−0.056
95% family bounds	[−0.016, 0.006]	[−0.089, −0.013]	[−0.099, −0.012]

Units are months of the category's pre-decision purchase flow per 25-basis-point intended-policy innovation. Each panel applies a separate nine-window Bonferroni adjustment. The full response equals the first-year and later response exactly. Samples contain 238 and 212 decisions, respectively; all windows within a panel share rows and controls.

policy history and leverage adjustment. The measured expenditure response is therefore not confined to consumer-durable purchases.

The category comparison still requires care. In the full sample, durable-minus-nondurable and durable-minus-services three-year contrasts have two-comparison family intervals containing zero. The larger durable point response alone does not establish a statistically distinct response. Outside the reserves-targeting origins, both contrasts are negative under that same adjustment: −0.283 months relative to nondurables and −0.284 relative to services. These are comparisons of responses normalised by different category-specific baseline flows, not contributions to an additive expenditure loss.

The pattern is consistent with rapid durable adjustment and more general spending effects accumulating later. It cannot establish that the same households postponed and subsequently completed purchases, or attribute the services response specifically to income rather than wealth or credit. The aggregate categories describe the scope of the response, but do not discriminate among timing, income, wealth or credit mechanisms.

4.3 Observed rate implementation

I check implementation using the separately observed daily effective federal funds rate. Before fitting outcomes, I fixed a financial window equal to the mean of seven published calendar-day quotes after each decision minus the mean of seven before it, excluding the decision day. All source-valid windows are complete and none overlaps another decision listed in the original source record. Published weekend and holiday repetitions enter unchanged. This window can still contain other news and intermeeting actions.

The full-sample rate response is 0.174 annual percentage points per 0.25-point intended-policy innovation, with approximate bounds [0.126, 0.222]. The corresponding response outside the reserves-targeting origins is 0.126 points, with bounds [0.083, 0.169]. Thus the recorded innovation has a measurable implementation response, although intentions and the immediate realised rate are different objects. Historical rate measurement changed in 1973; I retain the published convention rather than describe the whole sample as a transaction-median series.[1]

[1]The Federal Reserve's September 1973 Bulletin, p. A33, records the change from a representative daily transaction rate to a volume-weighted mean beginning with the statement week ending 25 July 1973. The modern median convention began in 2016, outside this sample.

For comparison I scale the purchase response to a 25-basis-point observed financial impact. If a_H is the purchase coefficient and f the rate coefficient,

$$\theta_H = 0.25\frac{a_H}{f}. \tag{5}$$

Joint Anderson–Rubin-style inversion gives the full-sample durable ratio −0.205 months with bounds [−0.420, −0.011], and the outside-episode ratio −0.675 with bounds [−1.218, −0.264]. The calculation includes uncertainty in the financial scale. It describes the identified policy innovation per unit of its rate impact; it does not turn the effective rate itself into an exogenous treatment or impose a fixed future rate path. The native intentions scale remains primary.

5 Robustness and the role of decision origins

5.1 Sensitivity to dynamics and covariance

Table 2 reports every declared primary check. The durable point response remains negative with six or twenty-four policy lags, different dependence bandwidths, leverage adjustment and inclusion of the partial decision month. Its pointwise interval excludes zero with six lags and both bandwidth changes, but includes zero with twenty-four lags and leverage-adjusted covariance. The larger uncertainty is material: it limits the precision of the pooled three-year durable response and cannot be dismissed as a cosmetic change in standard errors.

Table 2: Declared checks of three-year durable and services responses

Design	Durables	Durable 95% bounds	p	Services family 95%
Primary: twelve policy lags	−0.142	[−0.277, −0.007]	0.039	[−0.064, −0.011]
Six policy lags	−0.139	[−0.274, −0.005]	0.042	[−0.068, −0.005]
Twenty-four policy lags	−0.122	[−0.285, 0.042]	0.144	[−0.071, −0.001]
Shorter dependence bandwidth	−0.142	[−0.278, −0.007]	0.040	[−0.066, −0.009]
Longer dependence bandwidth	−0.142	[−0.276, −0.008]	0.037	[−0.063, −0.012]
Leverage-adjusted covariance	−0.142	[−0.308, 0.024]	0.093	[−0.072, −0.003]
Start in the decision month	−0.141	[−0.274, −0.009]	0.036	[−0.063, −0.010]
Omit reserves-targeting decisions	−0.340	[−0.548, −0.131]	0.001	[−0.099, −0.012]

All rows were fixed before US spending estimation. Units are months of pre-decision purchases per 25 basis points of intended policy. Durable bounds and p values are pointwise; services bounds use each design's nine-window family adjustment. No confidence family across all designs is claimed. Bandwidths are 35, 41 and 47 calendar months. Leverage adjustment divides event residuals by one minus their leverage before monthly score aggregation. The origin exclusion removes October 1979–October 1982, inclusively; follow-up and history observations can still cross that episode.

No single-decision deletion reverses the primary sign; the estimates range from −0.182 to −0.098 months. This is a point-estimate influence check, not a confidence statement over deletions. Maximum baseline leverage is 0.747, making the covariance sensitivity especially relevant. The predetermined twelve-month durable-purchase outcome has $p = 0.116$. That test does not reject, but it cannot verify the information assumption or rule out anticipation. I retain it as a diagnostic rather than certification of exogeneity.

5.2 An informed comparison of decision origins

The exclusion of October 1979–October 1982 decision origins was specified before spending estimation because intended-rate coding is less clear during nonborrowed-reserve targeting. The retained outside-episode sample contains 212 decisions. Its durable total is −0.340 months, with a nine-window family interval [−0.635, −0.045]; the later two years add −0.280 months, also negative under that adjustment. The period exclusion thus changes both magnitude and the evidence of additional shortfall after the first year.

Coibion (2012) shows that reserves-targeting observations and dynamic lag choices can materially affect narrative monetary estimates. My boundary is a declared diagnostic, not a replication of his later resumption date. Removing decision origins does not isolate conventional-regime transmission.[2]

After observing the larger secondary estimate, I recorded a bounded follow-up: cross the origin exclusion with leverage adjustment and with twenty-four policy lags, and test its coefficient difference from the retained primary. Both crossed pointwise intervals remain negative. The leverage-adjusted response is -0.340, with bounds $[-0.604, -0.076]$; the longer-history response is -0.311, with bounds $[-0.576, -0.046]$. Across the three informed diagnostics, the Bonferroni family interval remains negative for leverage adjustment but includes zero for the longer-history check. This follow-up strengthens the secondary evidence without replacing the primary or concealing the weaker family result.

The subset-minus-primary difference is -0.197 months. Its common-calendar covariance includes the overlap between the two estimated samples. The approximate 95 per cent interval is $[-0.340, -0.055]$ and its three-diagnostic family interval is $[-0.372, -0.023]$. The joint interval excludes zero in this informed comparison. The three-statistic adjustment does not correct the earlier data-dependent choice to investigate the difference, so this finding remains exploratory. It does not identify a causal effect of switching monetary regimes: changing the sample also changes policy variance, macroeconomic states, information quality and the weights in the linear projection.

The economic implication is therefore specific. The pooled estimate compresses a materially larger durable contraction found among decisions outside the difficult coding episode. A single pooled number is an incomplete description of the retained narrative evidence. This observation complements, rather than substitutes for, the more stable services result and the common negative durable point responses.

6 Conclusion

I find a negative three-year cumulative response of household purchases to an upward innovation in recorded US monetary-policy intentions. The declared durable estimate is a shortfall of 0.142 months of pre-decision purchases per 25 basis points; its pointwise interval excludes zero, but longer policy history and leverage adjustment weaken its precision. The prespecified origin exclusion gives a larger durable deficit and an exploratory joint sample comparison whose interval excludes zero. Services exhibit a smaller, consistently negative cumulative response across the declared checks. These are distinct findings, with different degrees of robustness.

The mechanism is clearest when purchase flows and their cumulative quantity are kept separate. Resumption of purchases cannot by itself repay an earlier deficit. In the estimated three-year window, later durable spending contributes further shortfall rather than an offset at the point estimates. This does not establish permanent expenditure loss or reject eventual durable-demand reversal. Causal interpretation rests on the narrative information and dynamic restrictions; the observed-rate scaling is an implementation comparison, not an externally imposed policy path. The evidence supports a finite-horizon household-demand assessment whose magnitude depends materially on which monetary decisions enter the sample.

[2]Earlier decisions can have thirty-six-month purchase follow-up inside the excluded episode; later decisions can have policy histories reaching back into it. The comparison conditions on the retained decision origins and does not delete all observations associated with that monetary regime.

A Purchase accounting and normalisation

Subtracting the no-shock stock path from equation (1) gives $\Delta X_j = \Delta D_j - (1-\delta)\Delta D_{j-1}$. Summation yields

$$\sum_{j=1}^{H} \Delta X_j = \Delta D_H - \Delta D_0 + \delta \sum_{j=1}^{H} \Delta D_{j-1}. \tag{6}$$

For an initially unaffected stock, the cumulative purchase deviation equals its endpoint deviation plus depreciation on the intervening stock deviations. An endpoint purchase response of zero imposes no restriction on this sum. An endpoint stock deviation of zero also need not make it zero. This is a standard accounting result for the illustrative stock law, not a claim of theoretical novelty or a literal reconstruction of the BEA stock.

Let A_i be category i's reference-year annual nominal expenditure. Its monthly real flow is $F_s^i = (A_i/12)Q_s^i/100$. Therefore $F_s^i/F_{t-1}^i = Q_s^i/Q_{t-1}^i$ exactly. The dollar anchor and annual-rate factor cancel. Summing within a category gives a real-flow ratio over time; summing unrelated chained category levels would not create a valid total quantity. Equation (3) estimates the former.

On identical rows and controls, the linearity of least squares gives $\widehat{\beta}_{36} = \widehat{\beta}_{12} + \widehat{\beta}_{13:36}$. The same identity holds for coefficient scores. The variance of the later-period estimate is $V_{36} + V_{12} - 2C_{36,12}$; separate confidence intervals cannot be subtracted as if independent. All reported windows use this common covariance. A recovery fraction would divide by an estimated initial loss and can be unstable near zero; I report the jointly estimated amounts instead.

B Matched score inference and financial scaling

Let T contain raw intended changes, F the complete source-purge design and $z = (T - F\widehat{\eta})/0.25$. Since every column of F lies in the conditioning span W, the residual-maker M_W satisfies

$$M_W z = M_W T/0.25, \qquad \widehat{\beta}_H = \frac{z' M_W Y_H}{z' M_W z}. \tag{7}$$

Changing $\widehat{\eta}$ cannot change the residualised regressor. The two augmented designs have the same column span, fitted values, residuals, leverage and coefficient loadings for the policy response. Their joint score covariance is consequently identical when rows, finite corrections and kernels are held fixed. This established projection result (Ding, 2021) does not remove errors in T or F, omitted information or instability in the response model.

For $X = [z, W]$, let ℓ_e be event e's loading for the first coefficient in $(X'X)^{-1}X'$ and $\widehat{u}_e$ the vector of outcome residuals. The event score is $s_e = \ell_e \widehat{u}_e$. Sum these within calendar month m, giving $g_m = \sum_{e:t(e)=m} s_e$, and embed zero scores in genuine months without selected decisions. Then

$$\widehat{V} = \frac{n}{n-k} \sum_m \sum_q K_L(m-q) g_m g_q', \qquad K_L(d) = \max\{0, 1 - |d|/(L+1)\}. \tag{8}$$

There is no additional division by n: the coefficient loadings already contain the regression scaling. Same-month cross-products are retained. The Bartlett kernel is positive semidefinite (Newey and West, 1987). Fixed $L = 41$ is the declared finite-sample rule; it alone does not supply a generic consistency argument. A slow-growing continuation $L(T) = 35 + \lceil 4(T/100)^{2/9} \rceil$ equals 41 at this calendar span and satisfies the required bandwidth growth condition, conditional on its moment and dependence restrictions. This is a theoretical clarification, not evidence of exact coverage or optimal bandwidth selection.

For two overlapping designs, I embed each design's scores on the full event calendar and multiply them by its own $\sqrt{n/(n-k)}$ before their joint covariance calculation. If a_S and a_F are the subset and full coefficients, the variance of their difference is

$$\widehat{\mathrm{Var}}(a_S - a_F) = V_{SS} + V_{FF} - 2V_{SF}. \tag{9}$$

The common-calendar calculation reproduces each design's marginal covariance. It tests an estimated sample coefficient difference, rather than treating one rejection and one non-rejection as evidence of different effects.

For equation (5), write $b = f/0.25$, $v_a = \widehat{\mathrm{Var}}(a)$, $v_b = \widehat{\mathrm{Var}}(b)$ and $c = \widehat{\mathrm{Cov}}(a, b)$. The approximate 95 per cent inverse set contains all θ satisfying

$$(a - \theta b)^2 \leq z^2_{.975}(v_a - 2\theta c + \theta^2 v_b). \tag{10}$$

I solve the quadratic without imposing boundedness. The inverse set can be bounded, unbounded or disconnected when the rate scale is weak. At $\theta = 0$, it gives the same zero-response test as the native purchase coefficient. This check does not confer an exclusion restriction on the observed rate. Its causal interpretation remains that of the underlying intended-policy innovation (Stock and Watson, 2018).

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